EEM 561 Machine Vision

Lecture 3
Light and Shading

Spring 2025

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Slides Credit: Frank (Qingzhong) Liu

Light and shading

A fine fall afternoon on Homer Lake

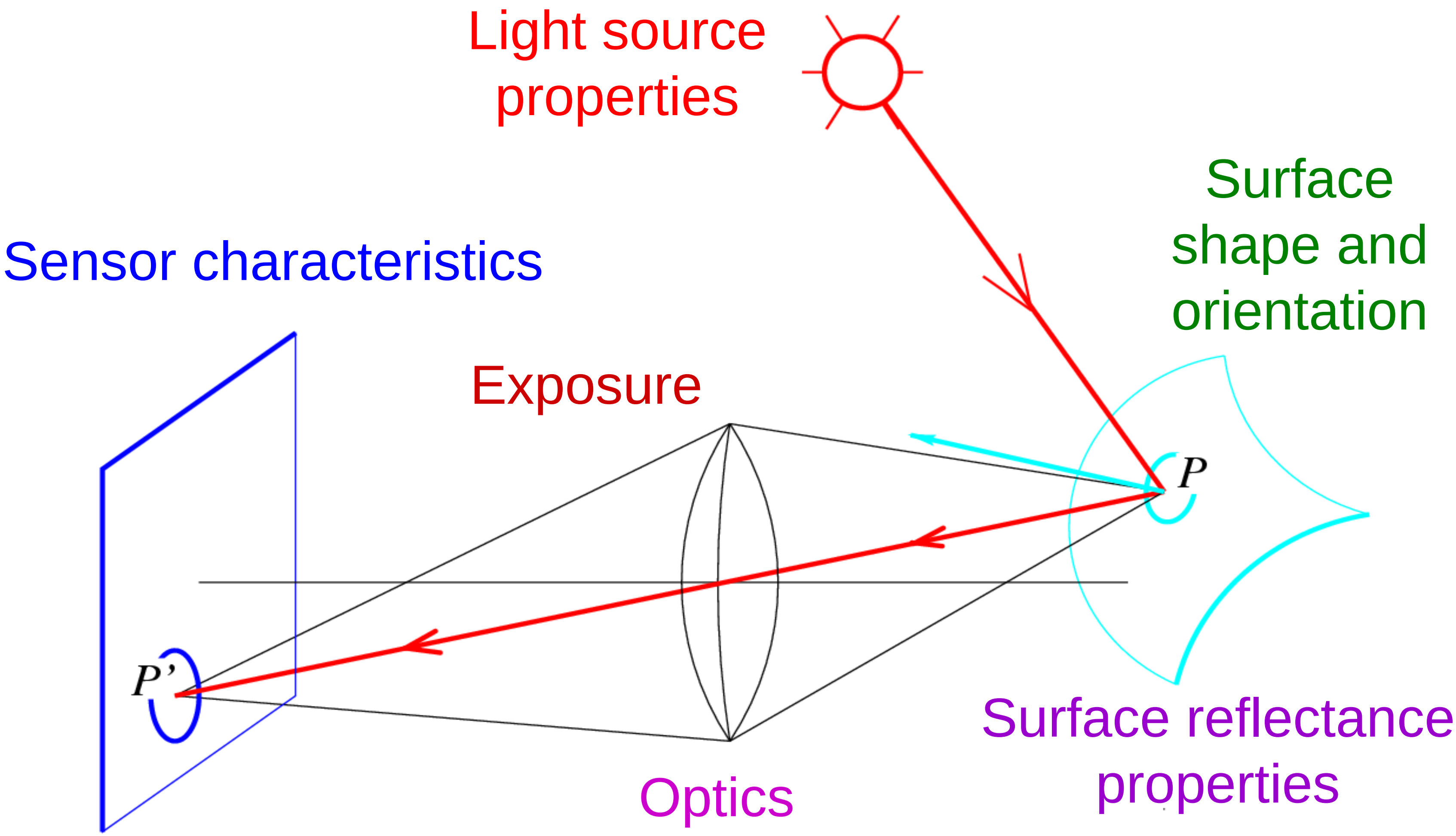
Light and shading

A fine fall afternoon on Homer Lake

Image formation

What determines the brightness of an image pixel?



Slide by L. Fei-Fei

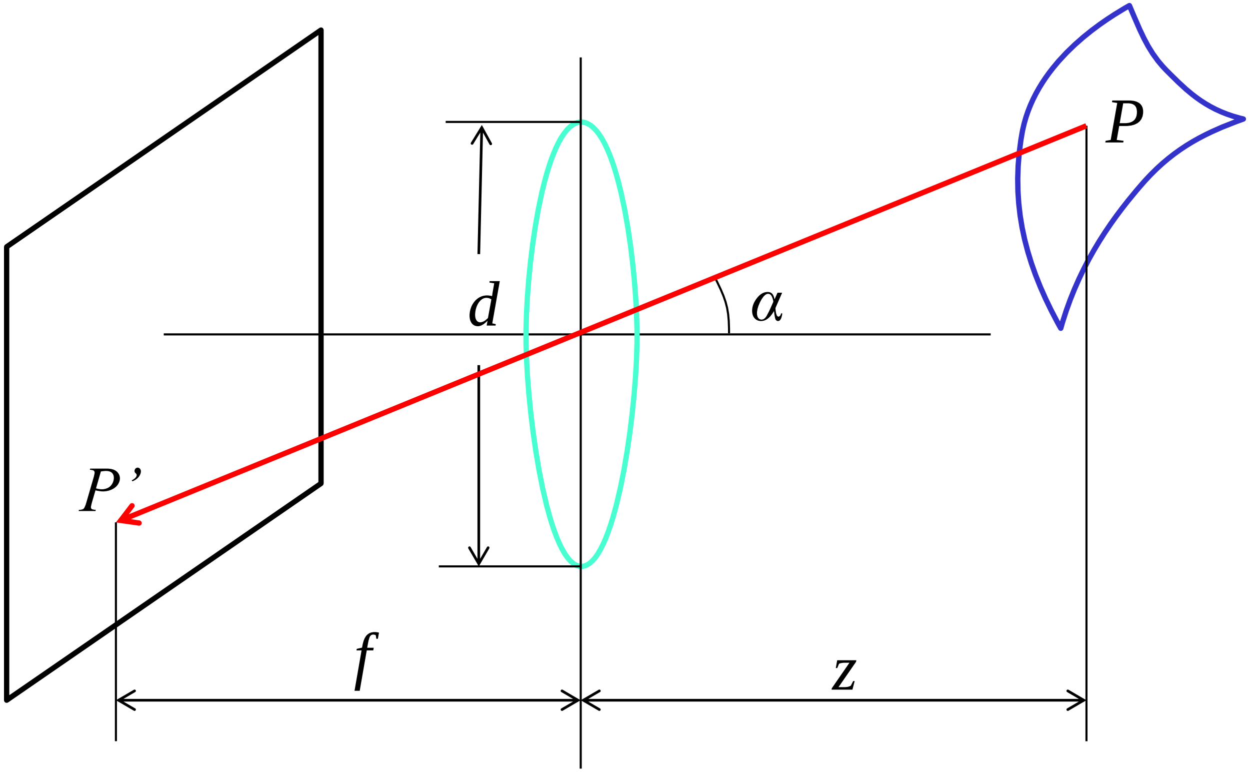
Fundamental radiometric relation

L: *Radiance* emitted from *P* toward *P*’

- Energy carried by a ray (Watts per sq. meter per steradian)

E: *Irradiance* falling on *P*’ from the lens

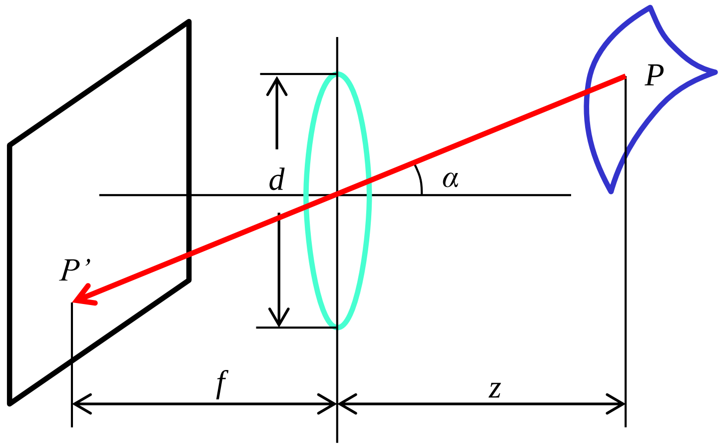
- Energy arriving at a surface (Watts per sq. meter)



What is the relationship between *E* and *L*?

Szeliski 2.2.3

Fundamental radiometric relation



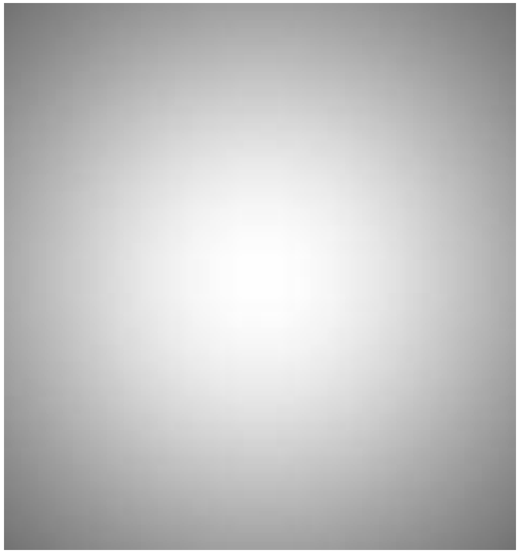
$$E = \left[\frac{\pi}{4} \left(\frac{d}{f} \right)^2 \cos^4 \alpha \right] L$$

- Image irradiance is linearly related to scene radiance
- Irradiance is proportional to the area of the lens and inversely proportional to the squared distance between the lens and the image plane
- The irradiance falls off as the angle between the viewing ray and the optical axis increases

Szeliski 2.2.3

Fundamental radiometric relation

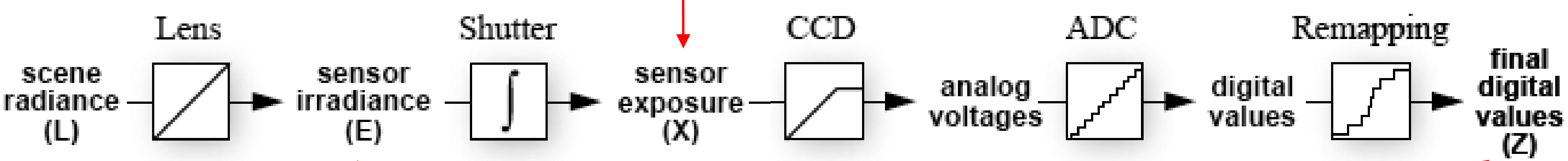
$$E = \left[\frac{\pi}{4} \left(\frac{d}{f} \right)^2 \cos^4 \alpha \right] L$$



S. B. Kang and R. Weiss, [Can we calibrate a camera using an image of a flat, textureless Lambertian surface?]() ECCV 2000

From light rays to pixel values

$$X = E \cdot \Delta t$$



$$E = \left[\frac{\pi}{4} \left(\frac{d}{f} \right)^2 \cos^4 \alpha \right] L$$

$$Z = f\left(E \cdot \Delta t\right)$$

- Camera response function: the mapping f from irradiance to pixel values
  - Useful if we want to estimate material properties
  - Enables us to create high dynamic range images
  - For more info: P. E. Debevec and J. Malik, *Recovering High Dynamic Range Radiance Maps from Photographs*, *SIGGRAPH 97*

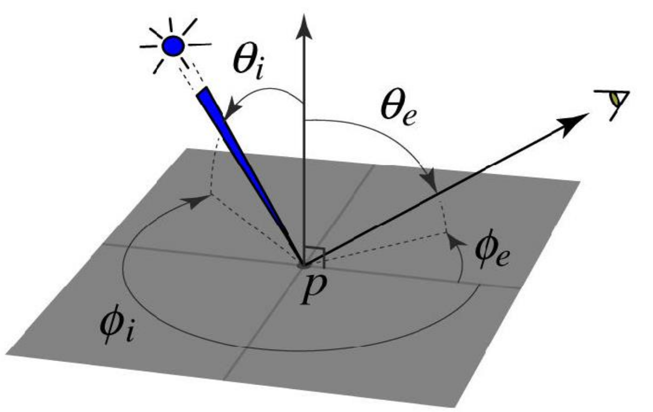
The interaction of light and surfaces

What happens when a light ray hits a point on an object?

- Some of the light gets **absorbed**
  - converted to other forms of energy (e.g., heat)
- Some gets **transmitted** through the object
  - possibly bent, through refraction
  - or scattered inside the object (subsurface scattering)
- Some gets **reflected**
  - possibly in multiple directions at once
- Really complicated things can happen
  - fluorescence

Bidirectional reflectance distribution function (BRDF)

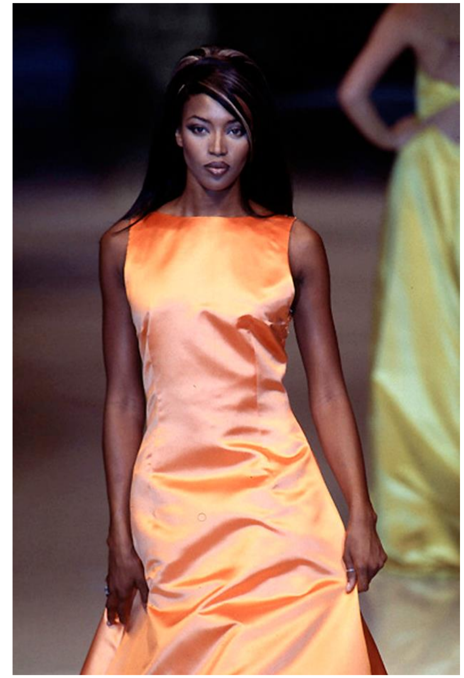
- How bright a surface appears when viewed from one direction when light falls on it from another
- Definition: ratio of the radiance in the emitted direction to irradiance in the incident direction



Source: Steve Seitz

BRDFs can be incredibly complicated…

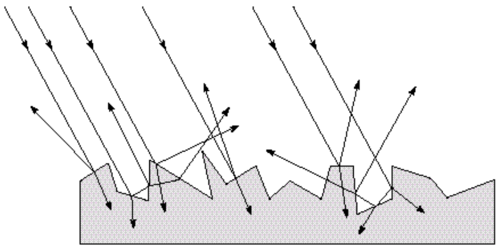


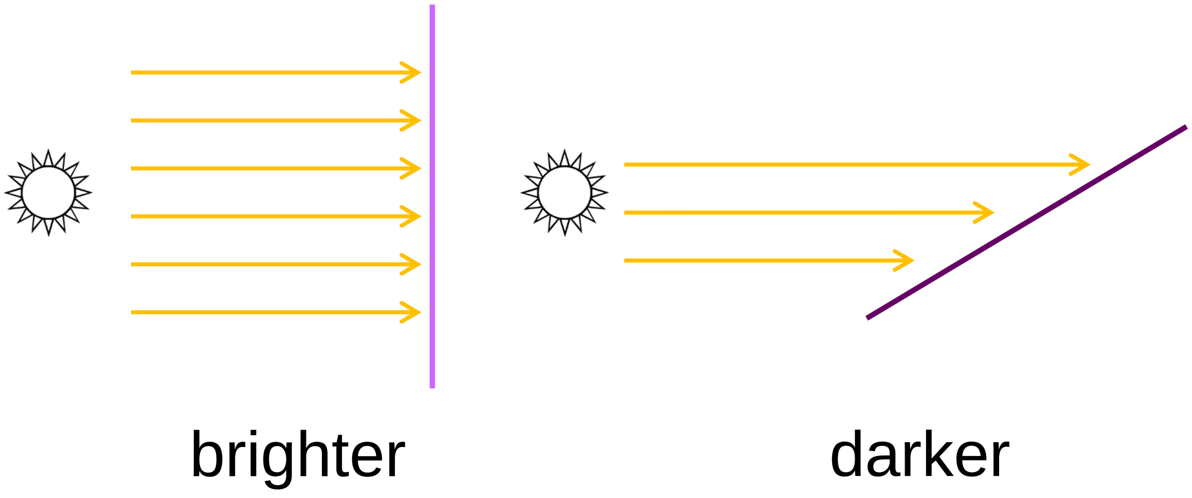




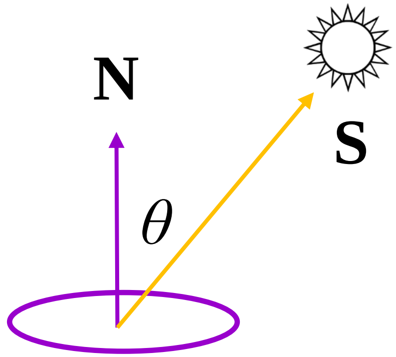

Diffuse reflection

- Light is reflected equally in all directions
  - Dull, matte surfaces like chalk or latex paint
  - Microfacets scatter incoming light randomly
- Brightness of the surface depends on the incidence of illumination



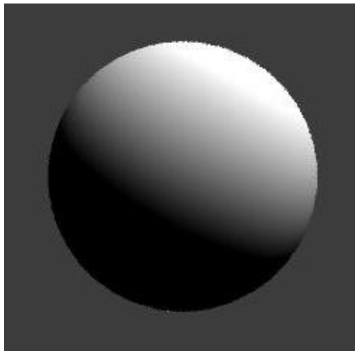


Diffuse reflection: Lambert's law



$$B = r\ \mathbf{N} \times \mathbf{S}$$

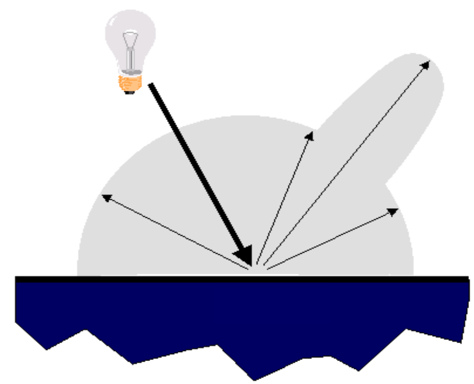
$$= r \left\|\mathbf{S}\right\| \cos q$$



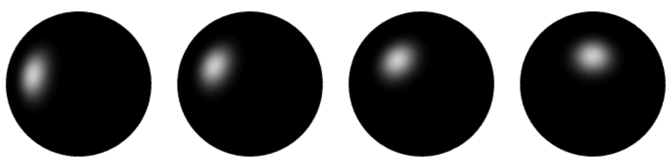
B: radiosity (total power leaving the surface per unit area)
ρ: albedo (fraction of incident irradiance reflected by the surface)
N: unit normal
S: source vector (magnitude proportional to intensity of the source)

Specular reflection

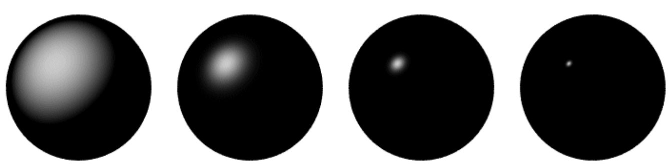
- Radiation arriving along a source direction leaves along the **specular direction** (source direction reflected about normal)
- On real surfaces, energy usually goes into a lobe of directions
- **Phong model**: reflected energy falls of with $\cos^n(\delta\theta)$

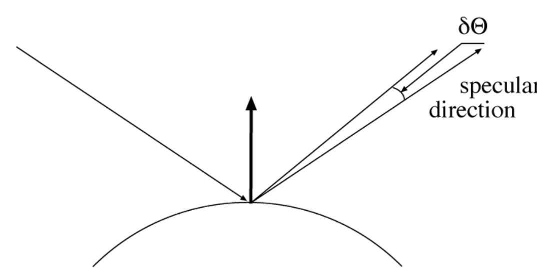


Moving the light source

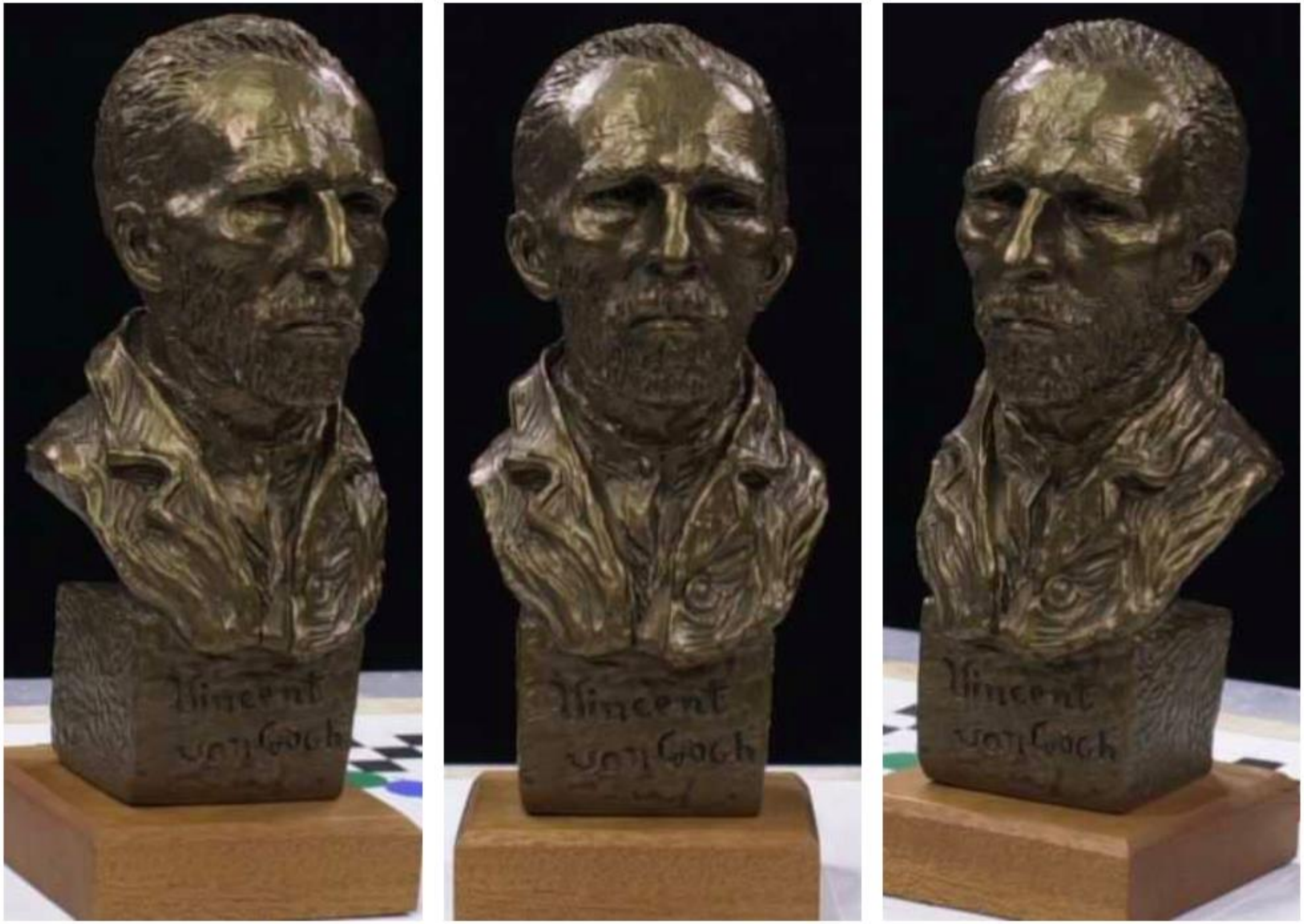


Changing the exponent





Specular reflection



[Picture source]

Photometric stereo (shape from shading)

- Can we reconstruct the shape of an object based on shading cues?

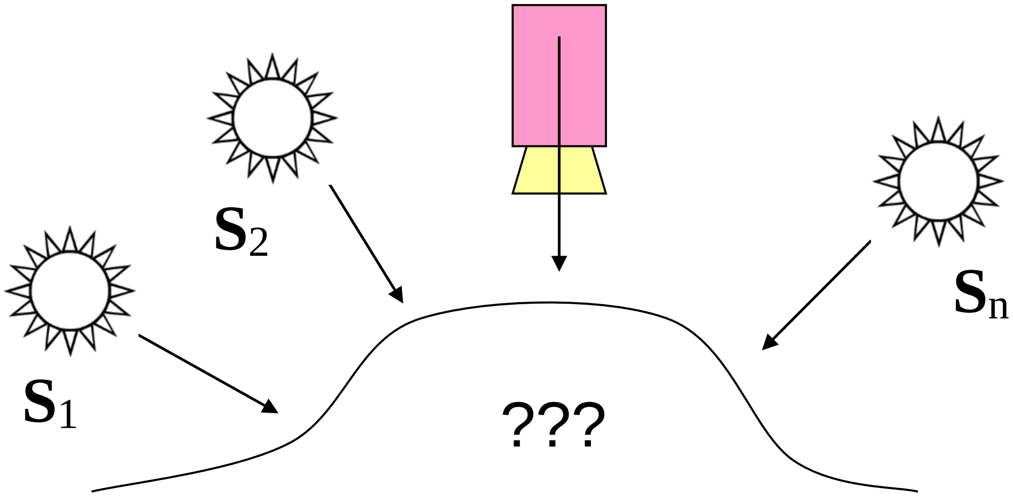
Luca della Robbia,
Cantoria, 1438

Photometric stereo

Assume:

- A Lambertian object
- A *local shading model* (each point on a surface receives light only from sources visible at that point)
- A set of *known* light source directions
- A set of pictures of an object, obtained in exactly the same camera/object configuration but using different sources
- Orthographic projection

Goal: reconstruct object shape and albedo



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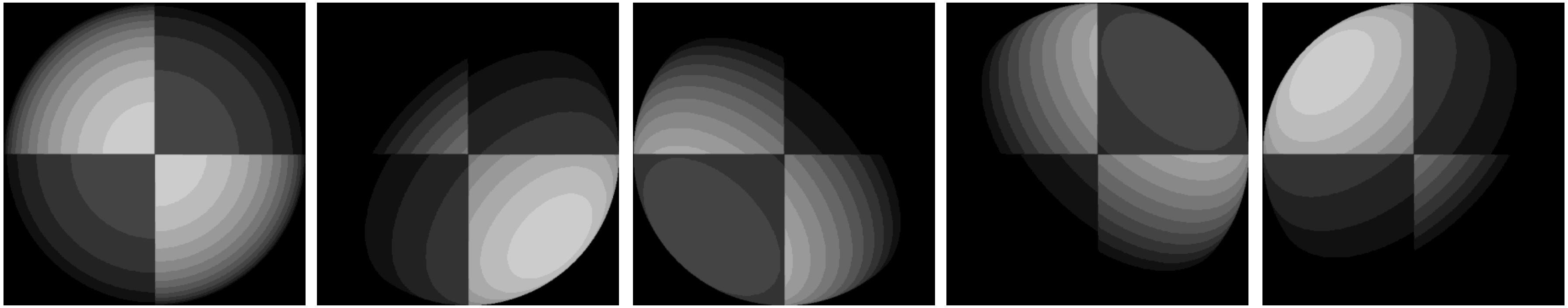
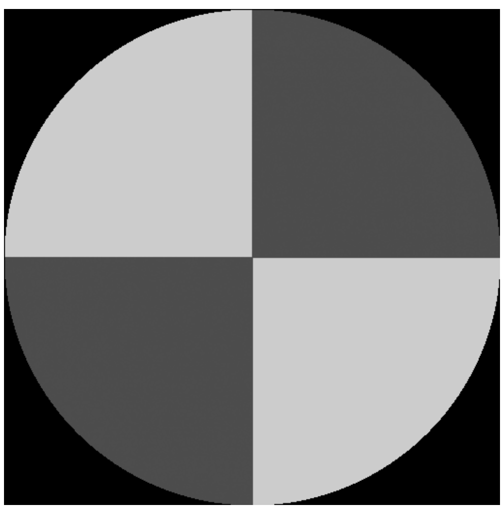
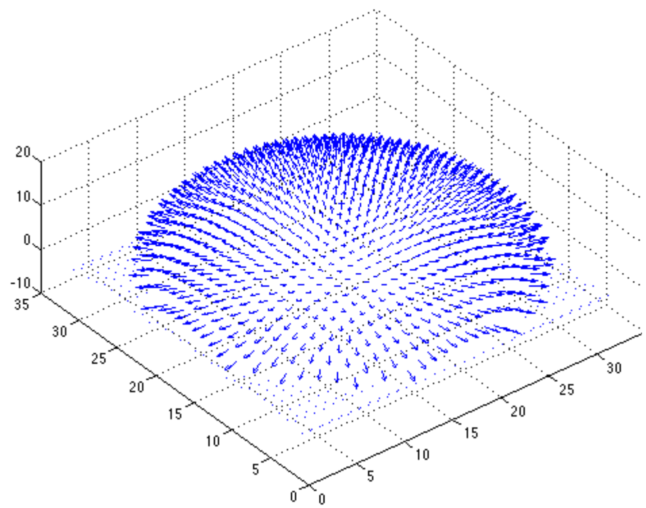
Example 1



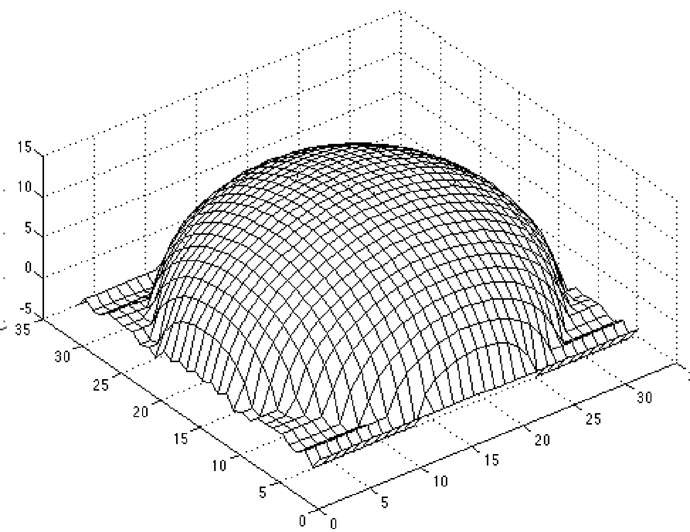
Recovered albedo



Recovered normal field



Recovered surface model



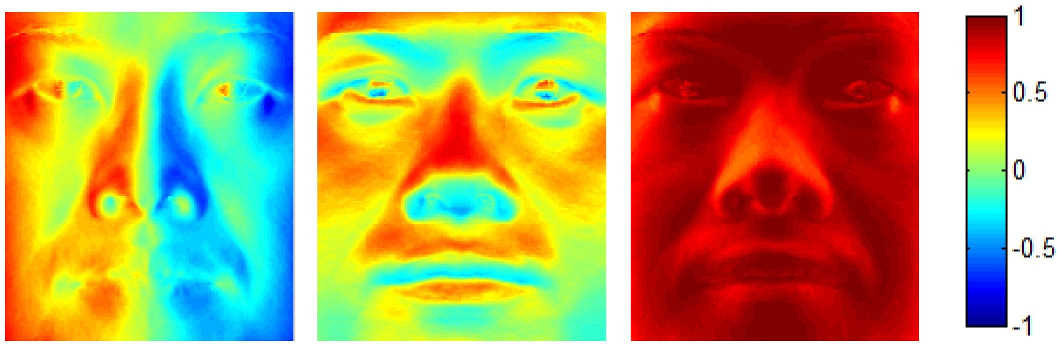
F&P 2nd ed., sec. 2.2.4

Example 2

Input

Recovered albedo

Recovered normal field



x y z

Recovered surface model

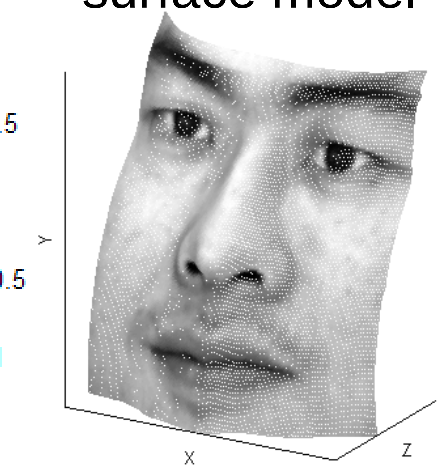


Image model

- **Known:** source vectors $\mathbf{S}_j$ and pixel values $I_j(x,y)$
- **Unknown:** surface normal $\mathbf{N}(x,y)$ and albedo $\rho(x,y)$

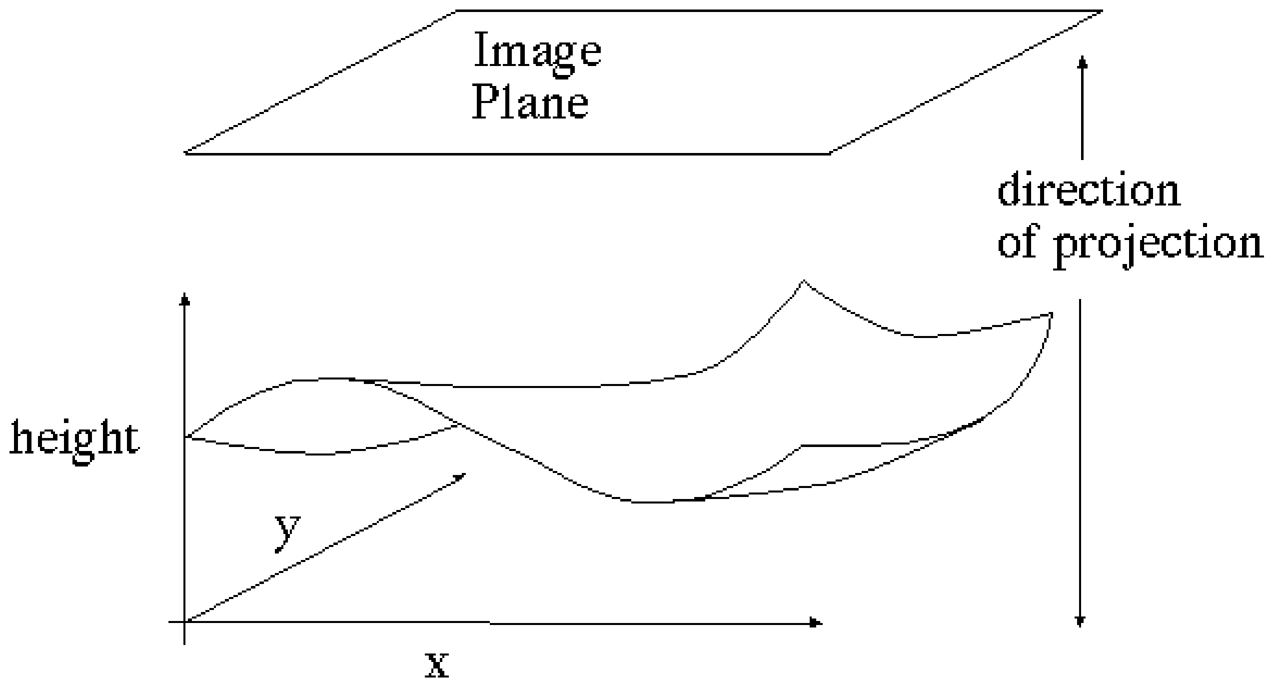


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Image model

- **Known:** source vectors $\mathbf{S}_j$ and pixel values $I_j(x,y)$
- **Unknown:** surface normal $\mathbf{N}(x,y)$ and albedo $\rho(x,y)$
- Assume that the response function of the camera is a linear scaling by a factor of k
- Lambert's law:

$$
\begin{aligned}
I_j(x,y) &= k\,\rho(x,y)\left(\mathbf{N}(x,y)\cdot\mathbf{S}_j\right) \\
&= \left(\rho(x,y)\mathbf{N}(x,y)\right)\cdot(k\mathbf{S}_j) \\
&= \mathbf{g}(x,y)\cdot\mathbf{V}_j
\end{aligned}
$$

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Least squares problem

- For each pixel, set up a linear system:

$$\underbrace{\begin{bmatrix} I_1(x,y) \\ I_2(x,y) \\ \vdots \\ I_n(x,y) \end{bmatrix}}_{\substack{(n \times 1) \\ \text{known}}} = \underbrace{\begin{bmatrix} \mathbf{V}_1^T \\ \mathbf{V}_2^T \\ \vdots \\ \mathbf{V}_n^T \end{bmatrix}}_{\substack{(n \times 3) \\ \text{known}}} \underbrace{\mathbf{g}(x,y)}_{\substack{(3 \times 1) \\ \text{unknown}}}$$

- Obtain least-squares solution for $\mathbf{g}(x,y)$ (which we defined as $\mathbf{N}(x,y)\ \rho(x,y)$)
- Since $\mathbf{N}(x,y)$ is the unit normal, $\rho(x,y)$ is given by the magnitude of $\mathbf{g}(x,y)$
- Finally, $\mathbf{N}(x,y) = \mathbf{g}(x,y) / \rho(x,y)$

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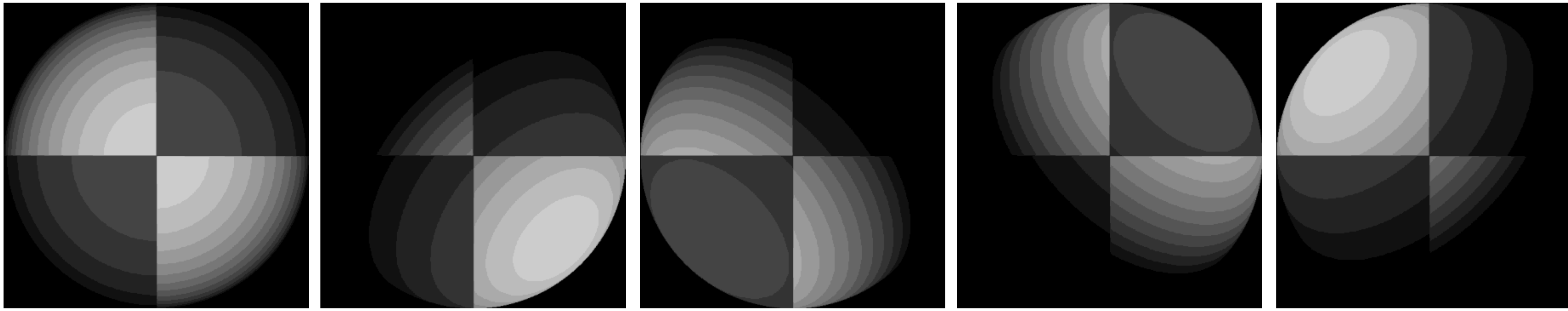
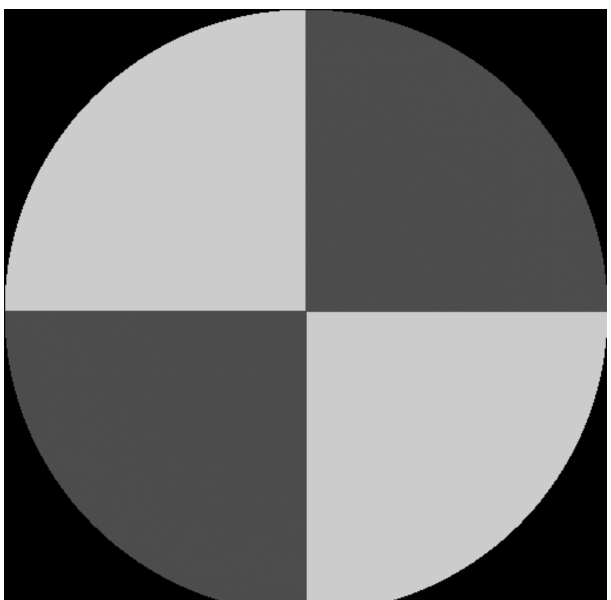
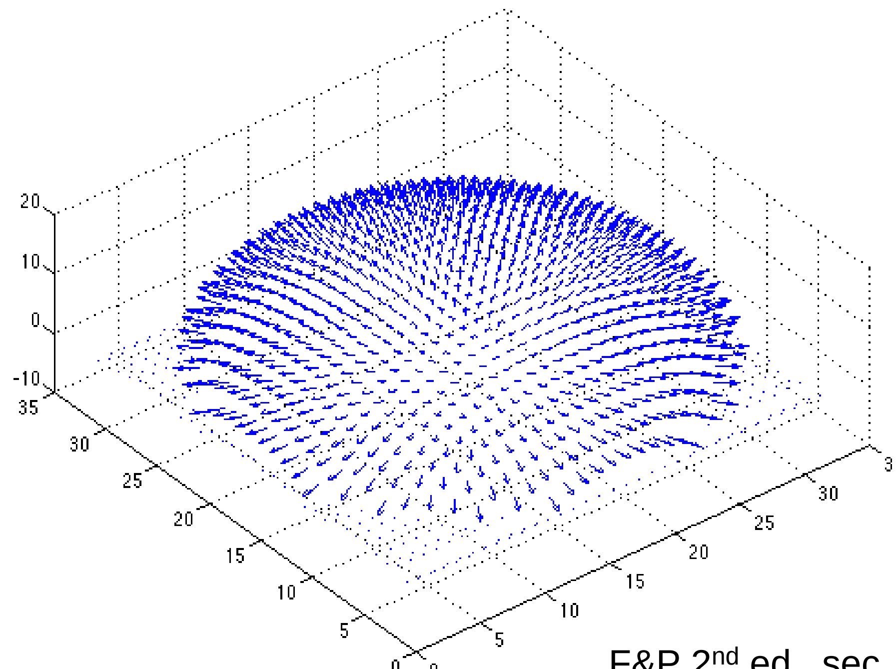
Synthetic example



Recovered albedo



Recovered normal field



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Recovering a surface from normals

Recall the surface is written as

$$(x, y, f(x, y))$$

This means the normal has the form:

$$\mathbf{N}(x, y) = \frac{1}{\sqrt{f_x^2 + f_y^2 + 1}} \begin{pmatrix} f_x \\ f_y \\ 1 \end{pmatrix}$$

If we write the estimated vector *g* as

$$\mathbf{g}(x, y) = \begin{pmatrix} g_1(x, y) \\ g_2(x, y) \\ g_3(x, y) \end{pmatrix}$$

Then we obtain values for the partial derivatives of the surface:

$$f_x(x, y) = g_1(x, y) / g_3(x, y)$$
$$f_y(x, y) = g_2(x, y) / g_3(x, y)$$

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Recovering a surface from normals

We can now recover the surface height at any point by integration along some path, e.g.

$$f(x,y) = \int_0^x f_x(s,0)\,ds + \int_0^y f_y(x,t)\,dt + C$$

(for robustness, should take integrals over many different paths and average the results)

Integrability: for the surface f to exist, the mixed second partial derivatives must be equal:

$$\frac{\partial}{\partial y}\left(g_1(x,y)/g_3(x,y)\right) = \frac{\partial}{\partial x}\left(g_2(x,y)/g_3(x,y)\right)$$

(in practice, they should at least be similar)

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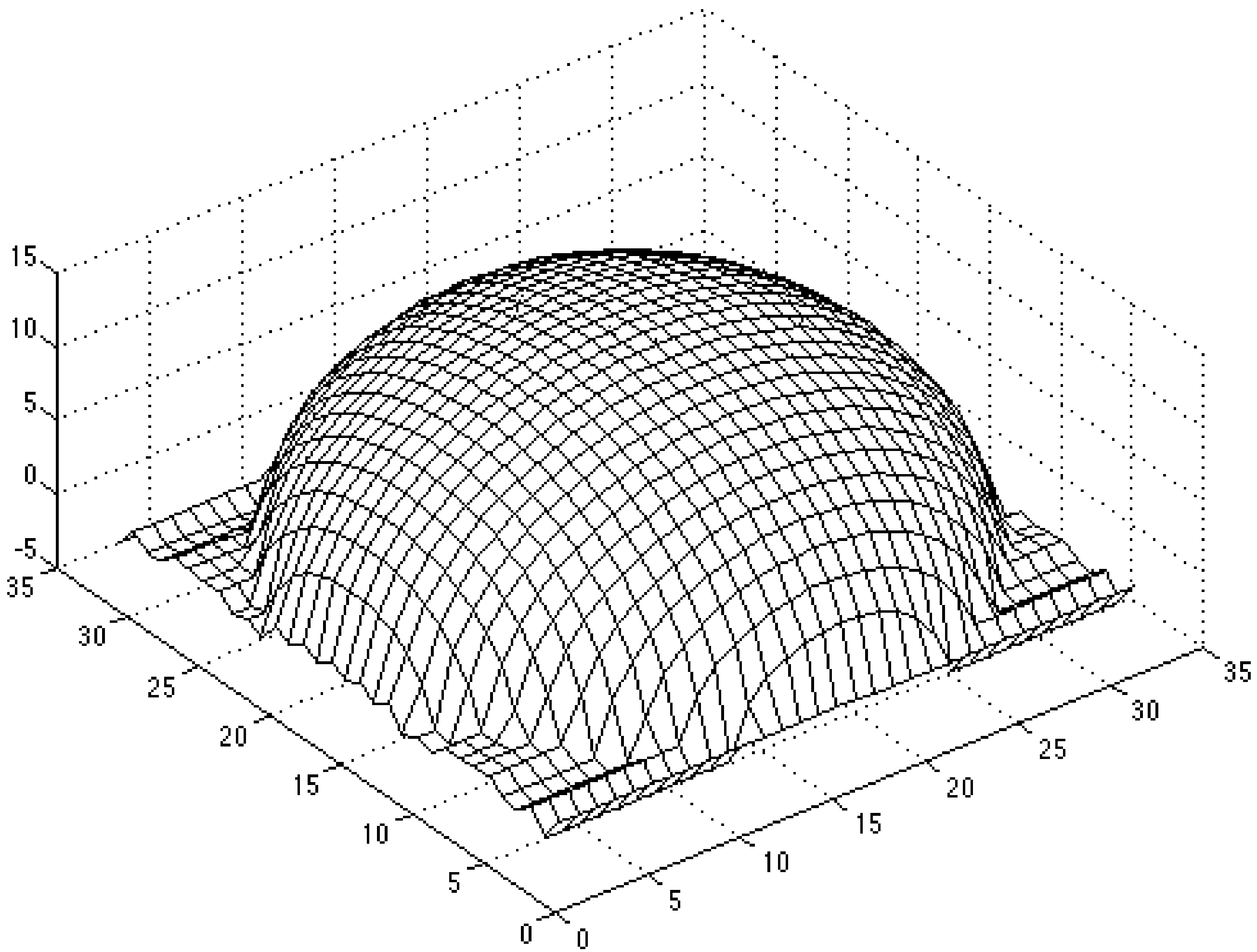
Surface recovered by integration



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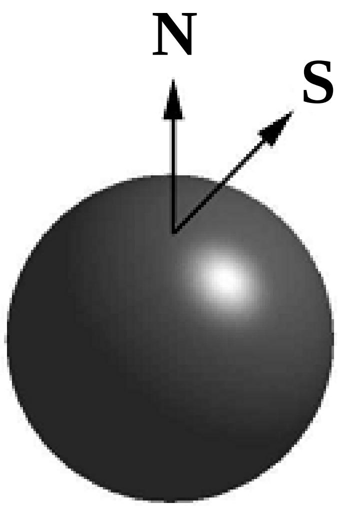
Limitations

- Orthographic camera model
- Simplistic reflectance and lighting model
- No shadows
- No interreflections
- No missing data
- Integration is tricky

Finding the direction of the light source

$$\boxed{I(x,y) = \mathbf{N}(x,y) \cdot \mathbf{S}(x,y)}$$



Full 3D case:

$$\begin{pmatrix} N_x(x_1,y_1) & N_y(x_1,y_1) & N_z(x_1,y_1) \\ N_x(x_2,y_2) & N_y(x_2,y_2) & N_z(x_2,y_2) \\ \vdots & \vdots & \vdots \\ N_x(x_n,y_n) & N_y(x_n,y_n) & N_z(x_n,y_n) \end{pmatrix} \begin{pmatrix} S_x \\ S_y \\ S_z \end{pmatrix} = \begin{pmatrix} I(x_1,y_1) \\ I(x_2,y_2) \\ \vdots \\ I(x_n,y_n) \end{pmatrix}$$

For points on the *occluding contour*:

$$\begin{pmatrix} N_x(x_1,y_1) & N_y(x_1,y_1) \\ N_x(x_2,y_2) & N_y(x_2,y_2) \\ \vdots & \vdots \\ N_x(x_n,y_n) & N_y(x_n,y_n) \end{pmatrix} \begin{pmatrix} S_x \\ S_y \end{pmatrix} = \begin{pmatrix} I(x_1,y_1) \\ I(x_2,y_2) \\ \vdots \\ I(x_n,y_n) \end{pmatrix}$$

P. Nillius and J.-O. Eklundh, “Automatic estimation of the projected light source direction,” CVPR 2001

Finding the direction of the light source

P. Nillius and J.-O. Eklundh, “Automatic estimation of the projected light source direction,” CVPR 2001

Application: Detecting composite photos

Real photo

Fake photo

M. K. Johnson and H. Farid, [Exposing Digital Forgeries by Detecting Inconsistencies in Lighting](), ACM Multimedia and Security Workshop, 2005.